

Springboard Virtual Internship 6.0

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Branch: Computer Science and Engineering

Domain: Data Visualization

Project: Visualizing U.S. Natural Disaster Declarations

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Date: 27-03-2026

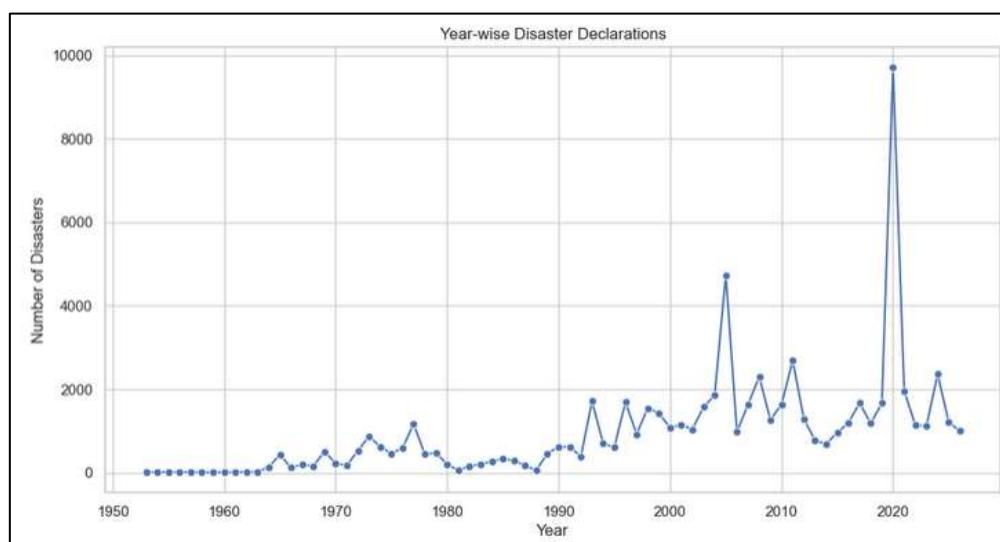
Insights Report-U.S. Natural Disaster Analysis

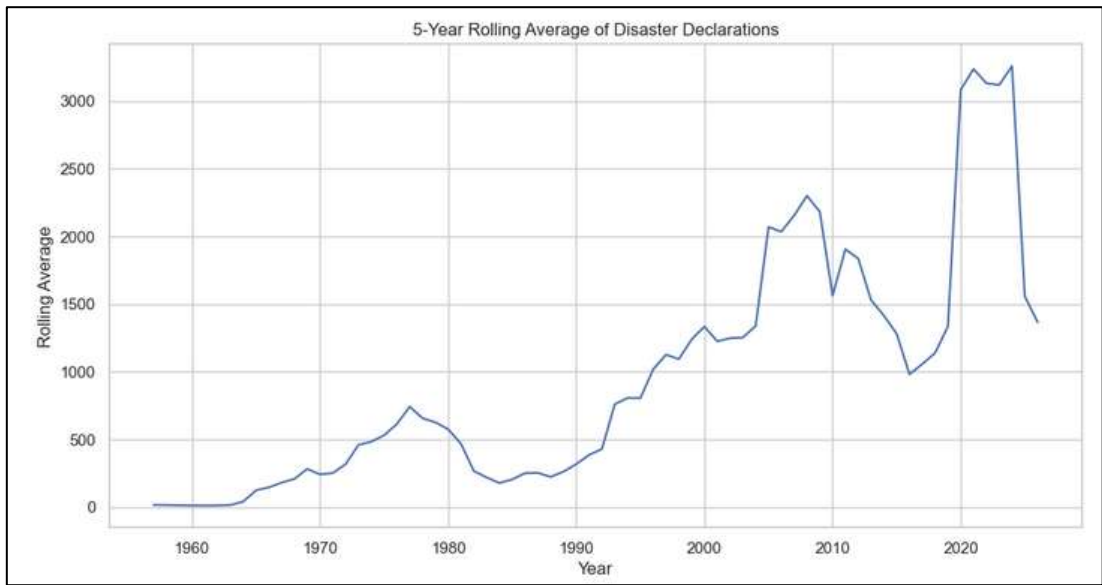
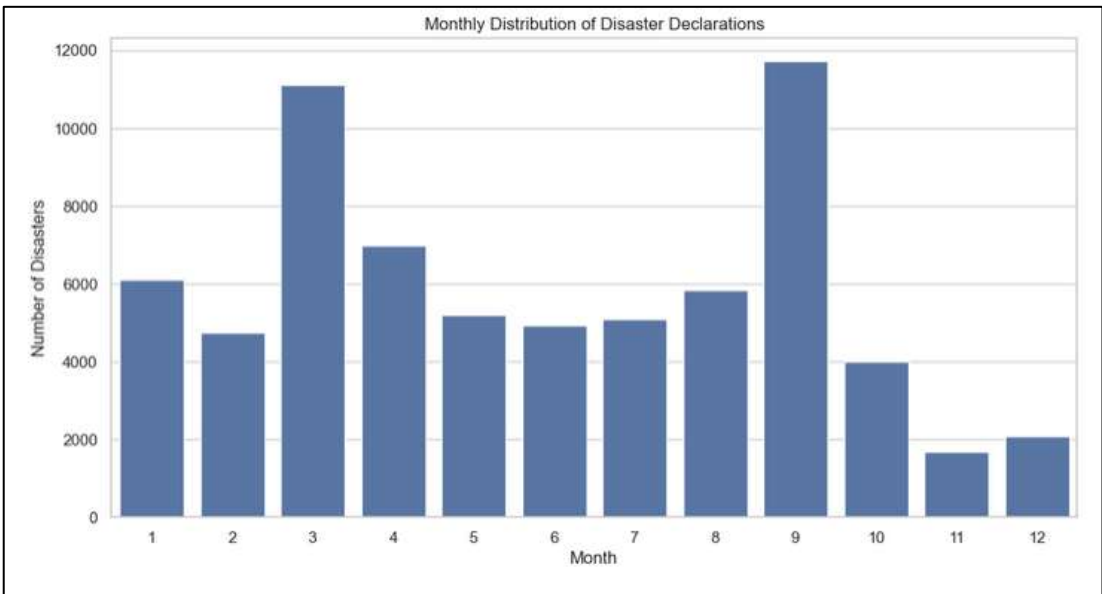
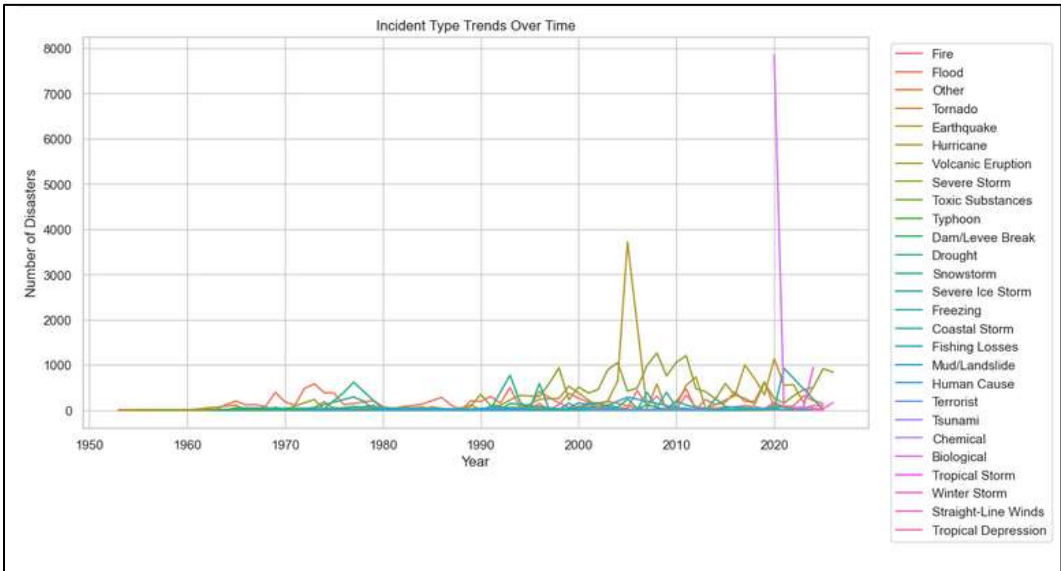
1. Overview of Insights

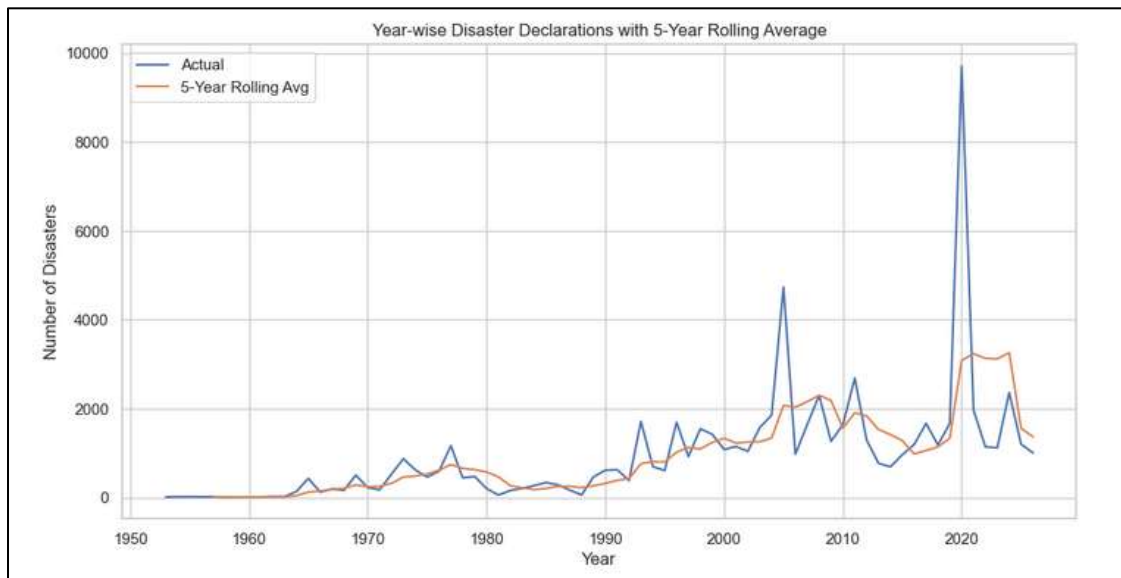
- This report summarizes the key insights derived from the analysis of U.S. natural disaster declarations.
- The insights are based on temporal, geographic, and incident type analysis performed across all milestones.
- The objective of this report is to identify patterns, trends, and relationships in disaster occurrences and provide a clear understanding of disaster behavior over time and across regions.

2. Temporal Analysis Insights

- Disaster declarations have shown a clear increasing trend over the years, indicating a rise in disaster occurrences or reporting.
- Certain months consistently show higher disaster frequency, highlighting strong seasonal patterns.
- The 5-year rolling average smoothens fluctuations and confirms long-term growth in disaster events.
- Some years show sharp spikes in disaster declarations, indicating extreme or multiple disaster events in those periods.
- The year-over-year growth rate fluctuates, showing that disasters are not evenly distributed and vary significantly year to year.

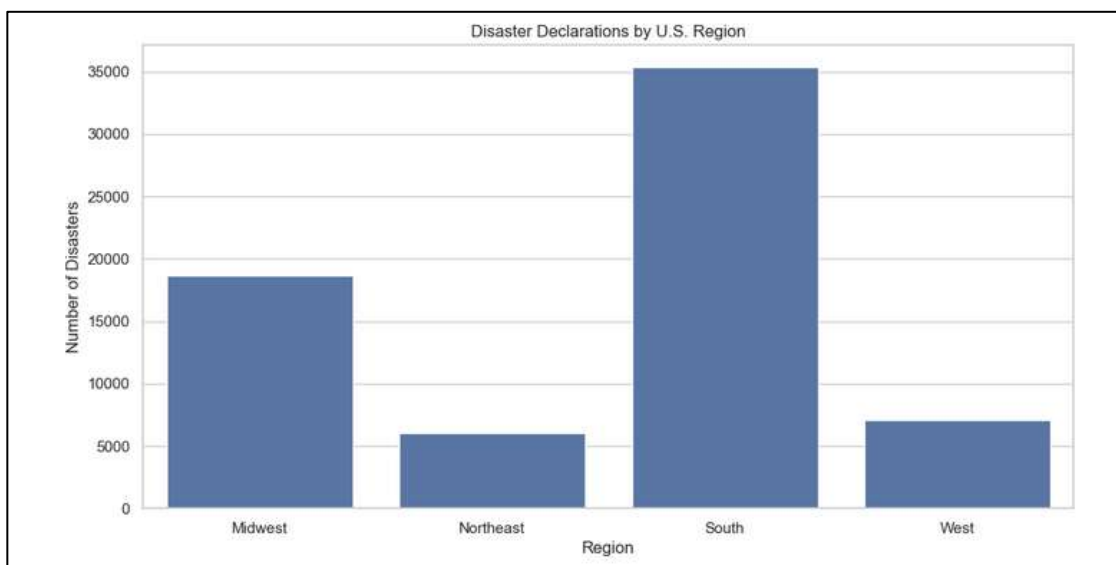




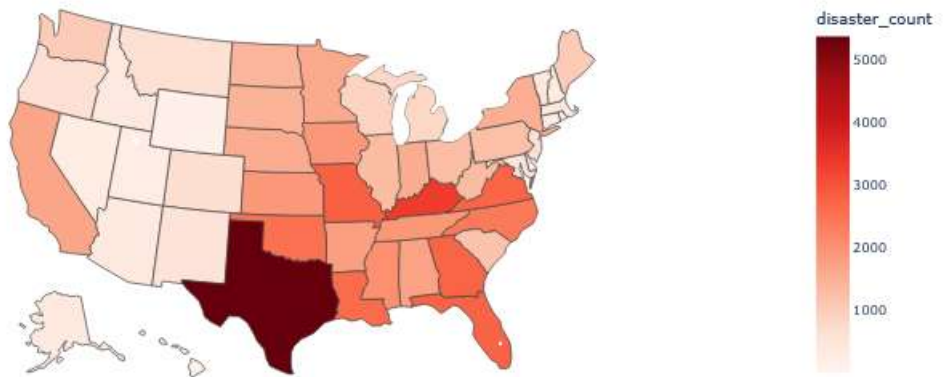


3. Geographic Analysis Insights

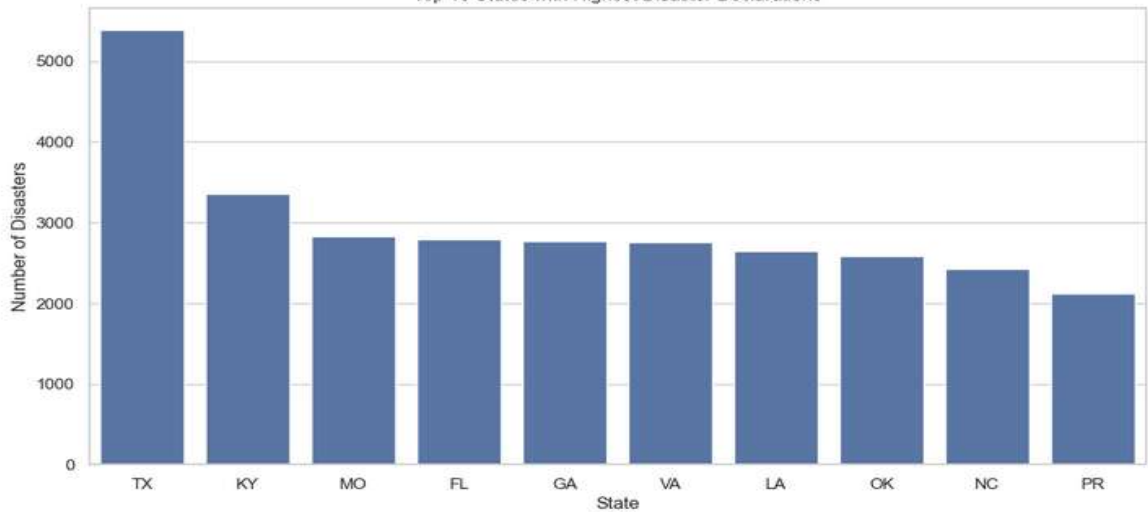
- The Southern region of the United States experiences the highest number of disasters, making it the most vulnerable region.
- Certain states consistently appear as high-risk zones, indicating repeated exposure to disasters.
- Hurricane-prone areas are concentrated along coastal regions, especially in the southeast.
- The choropleth map highlights that disasters are not evenly distributed geographically, with clear hotspots.
- Top affected states show a diverse mix of disaster types, indicating multiple vulnerabilities.



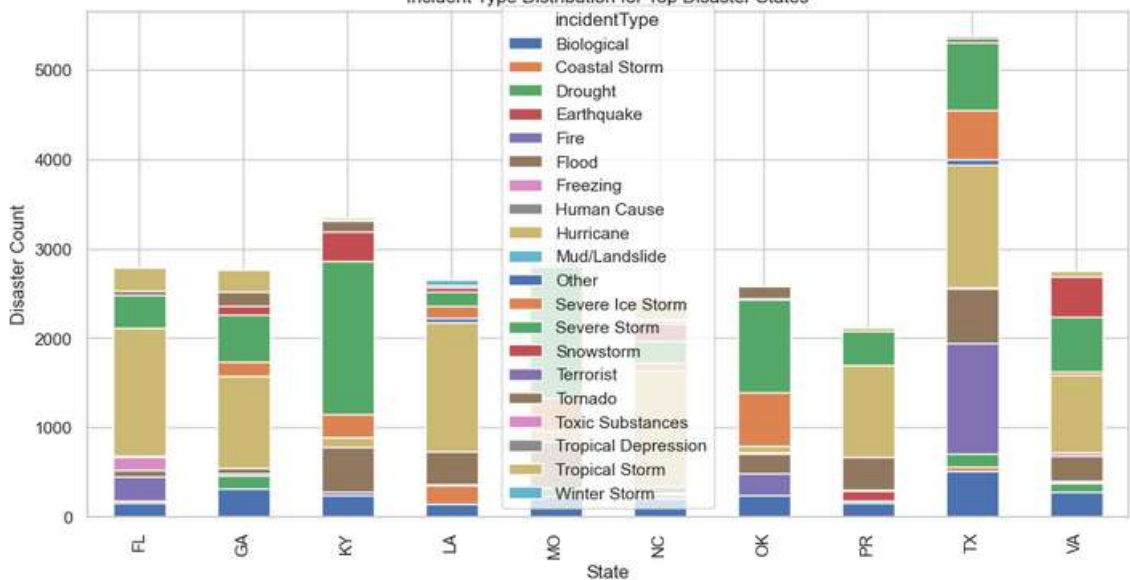
Disaster Declarations Across U.S. States



Top 10 States with Highest Disaster Declarations



Incident Type Distribution for Top Disaster States

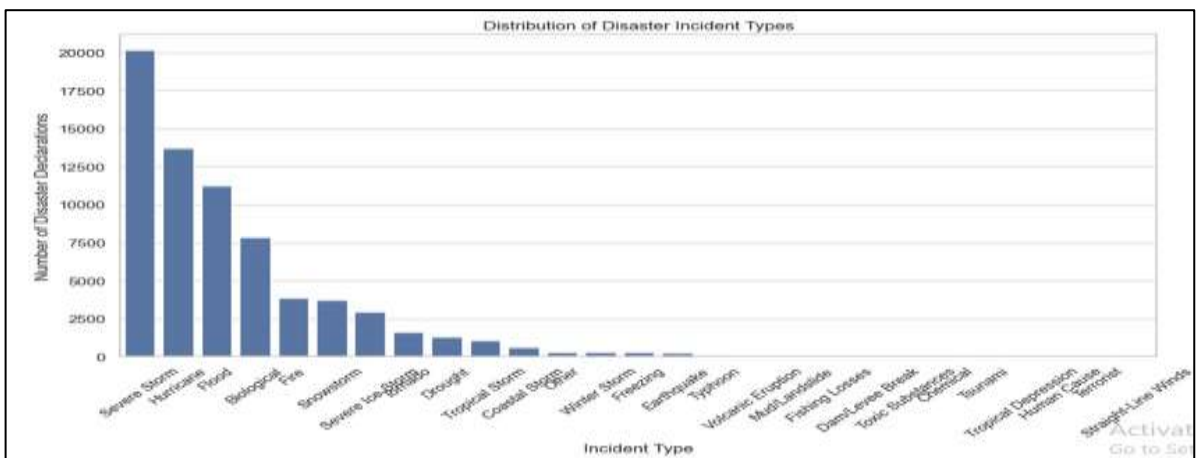


Hurricane Hotspots in the United States

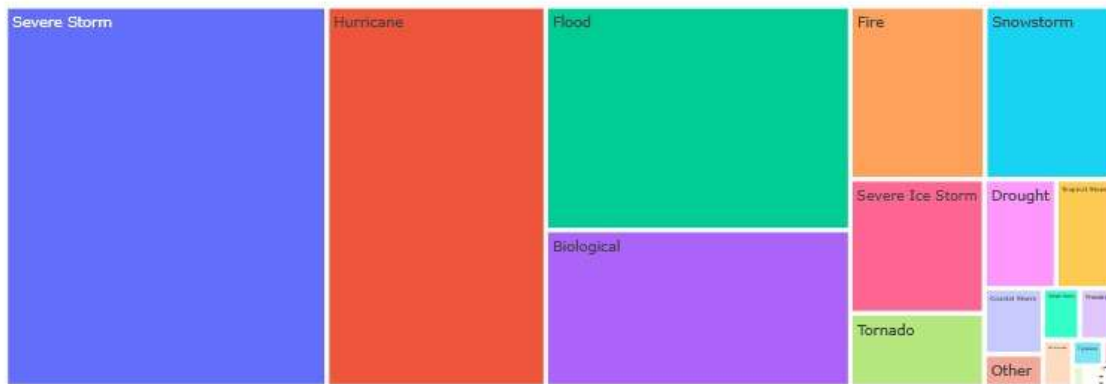


4. Incident Type Analysis Insights

- Some disaster types occur more frequently than others, showing dominance of specific incident categories.
- Weather-related disasters such as storms and floods are among the most common.
- Different states experience different dominant disaster types, showing regional variation.
- The distribution of disaster types is not uniform, indicating specialization of disaster patterns.
- Treemap and pie charts clearly show the proportion and contribution of each disaster type.

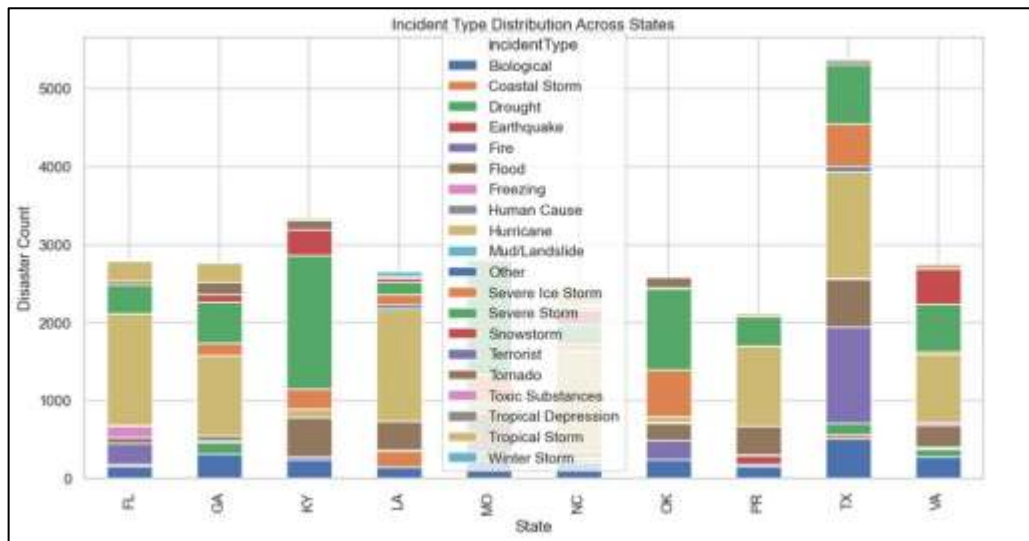


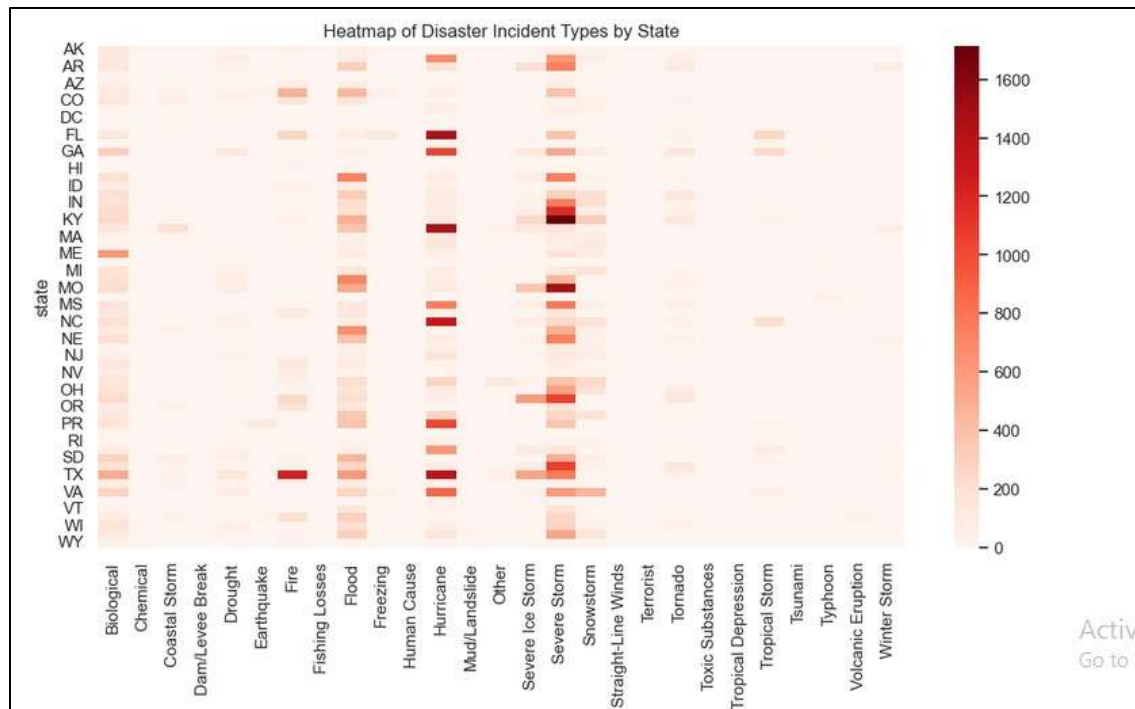
Treemap of Disaster Incident Types



5. State vs Incident Type Insights

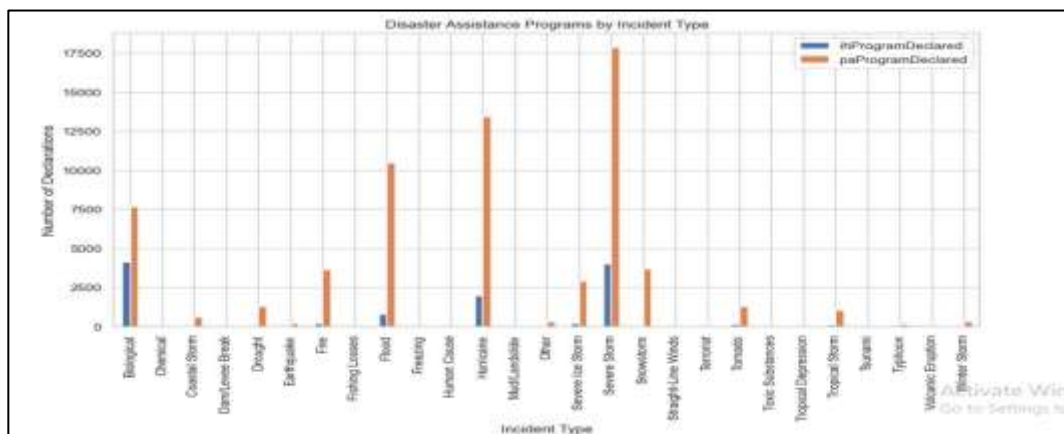
- Certain states are strongly associated with specific disaster types, such as hurricanes or floods.
- The heatmap reveals high-intensity clusters, indicating disaster-prone regions.
- Some states experience multiple disaster types, increasing their overall vulnerability.
- Regional patterns show that climate and geography influence disaster type occurrence.
- The stacked bar chart highlights variation in disaster composition across states.





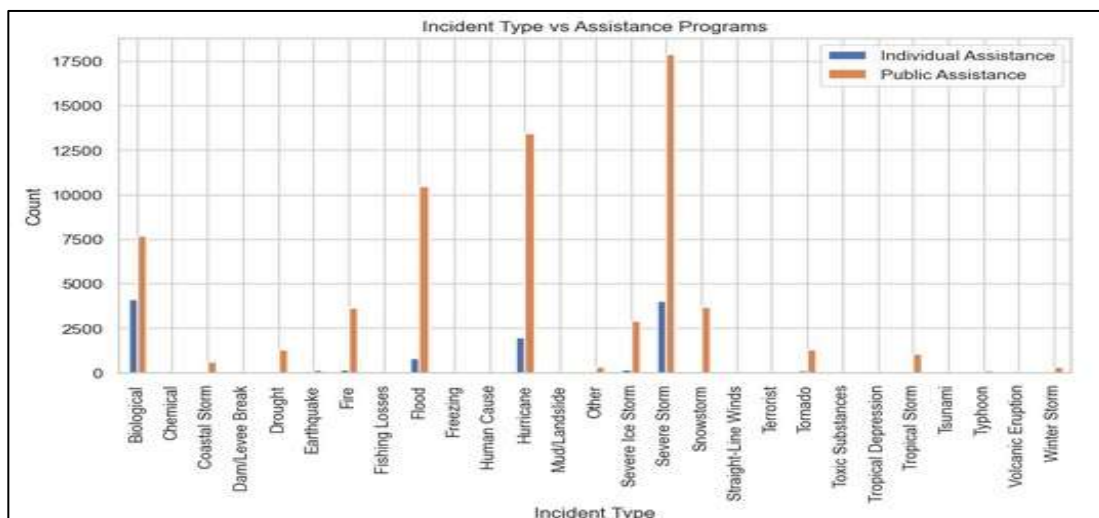
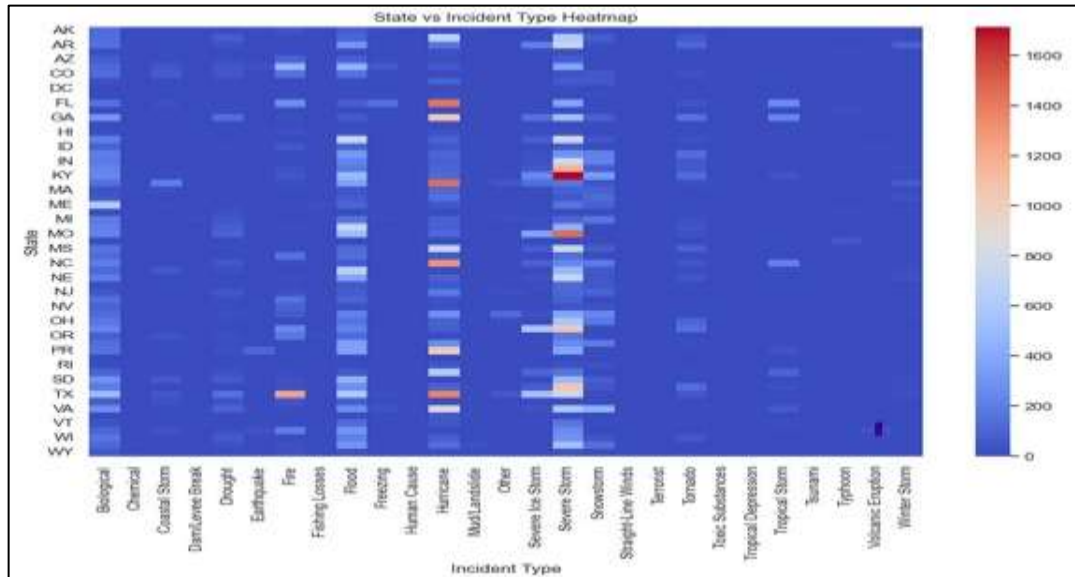
6. Disaster Assistance Program Insights

- Different disaster types trigger different assistance programs, depending on severity and impact.
- Some disasters require more public assistance, while others affect individuals more directly.
- There is a strong relationship between disaster type and assistance allocation.
- High-frequency disasters tend to receive consistent assistance support.
- The comparison shows how resources are distributed based on disaster impact.



7. Combined (Cross-Milestone) Insights

- Disaster frequency is increasing over time and concentrated in specific regions.
- Certain regions repeatedly experience specific types of disasters, forming clear patterns.
- Disaster types, locations, and assistance programs are strongly interconnected.
- High-risk states face both high frequency and diverse disaster types.
- The overall analysis shows that disaster behavior is predictable to some extent based on historical patterns.



8. Final Insights Summary

- Disasters are becoming more frequent over time.
- Geographic regions show clear vulnerability patterns.
- Certain disaster types dominate the dataset.
- Assistance programs are closely linked to disaster type and severity.
- Data visualization helps in better understanding and decision-making for disaster management.

9. Deployed Dashboard Link

- The interactive dashboard for this project has been successfully deployed and can be accessed using the link below:
- **Live Dashboard:** <https://u-s-natural-disaster-dashboard.vercel.app>